ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Data Science

CERTIFICATE

This is to certify that the Cornerstone project work entitled **STUDENT ATTENDANCE ANALYSIS** is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Data Science for the academic year 2024-2028.

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DECLARATION

We hereby declare that the Cornerstone project entitled "**STUDENT ATTENDANCE ANALYSIS**" is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree** in **DATA SCIENCE**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

Place: Surampalem
Date: 27-03-2026

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ABSTRACT

Data mining is an essential process used to extract meaningful patterns and valuable knowledge from large datasets. In the field of education, institutions generate vast amounts of data, particularly related to student attendance, which can be effectively analyzed to gain insights into student behavior and performance. This project focuses on analyzing student attendance data using data mining techniques implemented through Weka, a widely used machine learning and data analysis tool. The attendance data is collected from institutional sources and undergoes preprocessing steps such as data cleaning, handling missing values, and eliminating inconsistencies to ensure data accuracy and reliability. Once the data is preprocessed, it is imported into Weka, where various data mining algorithms are applied to discover hidden patterns and trends. Classification techniques like decision tree algorithms (J48) are used to categorize students based on attendance levels, while clustering methods such as K-Means help in grouping students with similar attendance behavior. Additionally, association rule mining is applied to identify relationships between attendance and other influencing factors. The system effectively detects students with low attendance, enabling early intervention by faculty and administration. Furthermore, Weka's visualization tools are utilized to present the analyzed results through graphs and charts, making the information easy to understand and interpret. The overall analysis supports educational institutions in improving attendance management systems, enhancing student engagement, and making informed decisions based on data-driven insights.

Keywords: Data Mining, Student Attendance, Weka, Data Preprocessing, Classification, Clustering, J48 Algorithm, K-Means, Association Rules, Educational Data Mining, Visualization

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1 INTRODUCTION

1.1 Introduction

In the digital era, educational institutions generate large amounts of data, but much of it remains underused. Data mining helps extract useful insights to improve academic decision-making.

Student performance is influenced by factors like attendance, study habits, and participation, which are not fully captured by exam results alone. This project combines performance data (marks, grades) and activity data (attendance, study hours) for better analysis.

After preprocessing the data using Python, data mining techniques are applied in Weka. Classification methods (Decision Tree, Random Forest) predict performance, while clustering (K-Means, Hierarchical) groups students.

The goal is to identify key factors affecting performance and help educators detect at-risk students, enabling timely interventions and improving educational outcomes.

1.2 Problem Statement

In the current education system, large volumes of student data such as academic performance, attendance, study habits, and participation are generated regularly. However, most institutions still rely on traditional evaluation methods that mainly focus on examination results, which do not provide a complete understanding of a student's overall performance and learning behavior. As a result, several important factors influencing academic success remain unidentified.

Another challenge is that student data is often stored in separate datasets, such as performance and activity data, making it difficult to integrate and analyze them effectively. Additionally, raw data may contain missing values, inconsistencies, and noise, which can affect the accuracy and reliability of analysis.

Therefore, there is a need for an efficient system that can integrate multiple datasets, preprocess data to improve quality, and apply data mining techniques to uncover meaningful patterns. This project aims to develop a data-driven approach to analyze student performance, identify key influencing factors, and improve prediction accuracy to support better decision-making in educational institutions.

1.3 Objectives of the Project

In the current education system, large amounts of student-related data such as academic performance, attendance, study habits, and participation are generated regularly. However, most institutions still rely on traditional evaluation methods focused mainly on examination results, which do not provide a complete understanding of a student's overall performance and learning behavior. As a result, many important factors influencing academic success remain unidentified.

Another issue is that student data is often stored in separate datasets, such as performance and attendance data, making it difficult to integrate and analyze them effectively. Additionally, the collected data may contain missing values, inconsistencies, and noise, which can reduce the accuracy and reliability of analysis.

Therefore, there is a need for an efficient system that can integrate multiple datasets, preprocess the data to improve quality, and apply data mining techniques to uncover meaningful patterns. This system should be capable of predicting student performance and identifying key factors such as attendance, study hours, and participation. The goal of this project is to develop a data-driven approach that enhances prediction accuracy and supports effective decision-making in educational institutions.

1.4 Scope of the Project

This project focuses on analyzing student performance using data mining techniques by integrating both student performance and student activity datasets. The scope of the project includes collecting relevant data, preprocessing it to ensure quality, and applying various data mining methods to extract meaningful insights from the data.

The study considers key factors such as marks, attendance, name, student id, class and section. By combining both academic and behavioral data, the project provides a more comprehensive understanding of student performance. The analysis is carried out using Python for data preprocessing and WEKA for implementing data mining algorithms.

The project covers the application of different data mining techniques, including classification, clustering, and association rule mining. Classification models are used to predict student performance, clustering techniques are used to group students based on similar characteristics, and association rule mining is applied to identify relationships between factors affecting performance.

The scope of the project is limited to the datasets used and the selected algorithms such as Decision Tree (J48), Random Forest, K-Means, Hierarchical Clustering, The results are based on historical data and may vary when applied to different datasets or real-time scenarios. This project does not include real-time system deployment but focuses on data analysis and model development.

Overall, the project provides a strong foundation for applying data mining techniques in the educational domain. It can be further extended to include real-time data analysis, advanced machine learning models, and practical system implementation in the future.

2 SYSTEM OVERVIEW

2.1 Existing System

In the current education system, student performance is mainly evaluated using traditional methods such as exams and assignments, with limited consideration of attendance and other behavioral factors. This provides only a partial view of a student's overall performance.

Student data, especially attendance and academic records, is often stored in separate datasets, making integrated analysis difficult. The analysis is usually manual or done using basic tools, which is time-consuming and prone to errors.

Additionally, existing systems do not use advanced techniques like data mining to analyze attendance patterns or predict student performance. Data quality issues such as missing or inconsistent attendance records are also not handled effectively.

Overall, the existing system lacks data integration, automation, and advanced analysis, making it inefficient for generating meaningful insights from student attendance data.

2.2 Limitations of Existing System

The existing system for managing student attendance has several limitations that reduce its effectiveness:

- It mainly records attendance (present/absent) without analyzing patterns or its impact on academic performance.
- Attendance data is stored separately from academic records, making integration and comprehensive analysis difficult.
- Analysis is mostly manual or done using basic tools, leading to time consumption and human errors.
- Advanced techniques like data mining or machine learning are not used to identify patterns or predict student behavior.
- Data quality issues such as missing or inconsistent records are not properly handled.
- The system cannot automatically identify students with low attendance or those at risk.
- It lacks real-time monitoring and continuous tracking of attendance trends.

- Limited insights and visualizations are available for better understanding.

Overall, the system lacks automation, integration, and advanced analysis, highlighting the need for a data mining-based solution.

2.3 Proposed System

To overcome the limitations of the existing system, this project proposes a data-driven approach for analyzing student attendance using data mining techniques. The system focuses on identifying patterns, trends, and factors that influence student attendance and performance.

Attendance data is collected and integrated with related student activity data using a common identifier to form a unified dataset. Data preprocessing is performed using Python (Pandas, Matplotlib) to handle missing values, remove inconsistencies, and normalize the data for better analysis.

Data visualization techniques are used to understand attendance patterns and identify irregularities. The system then applies data mining algorithms using WEKA. Classification methods such as Decision Tree (J48) and Random Forest are used to predict attendance categories, while clustering techniques like K-Means and Hierarchical Clustering group students based on similar patterns.

Model performance is evaluated using metrics such as accuracy, precision, and recall to select the best model.

Overall, the proposed system provides an automated and efficient framework for analyzing attendance data, identifying low-attendance students, and supporting data-driven decision-making in educational institutions.

2.4 Software Requirements

The implementation of the student attendance analysis project requires the following tools and technologies:

- Operating System: Windows 10 or higher (or Linux/macOS)
- Programming Language: Python 3.x (used for data preprocessing, cleaning, and visualization)
- Libraries:

  - Pandas – data manipulation and preprocessing
  - NumPy – numerical computations
  - Matplotlib – data visualization
  - Seaborn (optional) – advanced visualizations
- Data Mining Tool: WEKA – for applying classification and clustering algorithms
- Development Environment:
  - Jupyter Notebook or Google Colab – for coding and preprocessing
  - Anaconda (optional) – environment and package management
- Data Format: CSV – for storing and transferring data
- Documentation Tools:
  - Microsoft Word – documentation
  - Microsoft PowerPoint – presentations

These tools provide a complete environment for preprocessing, analyzing, and visualizing student attendance data.

2.5 Hardware Requirements

The project requires minimal hardware as it focuses on data preprocessing, visualization, and analysis using Python and WEKA. A standard computer system is sufficient.

- Processor: Intel Core i3 or higher (i5/i7 recommended)
- RAM: Minimum 4 GB (8 GB recommended for better performance)
- Storage: Minimum 250 GB (SSD preferred for faster processing)
- System Type: 64-bit operating system
- Input Devices: Keyboard, Mouse
- Output Devices: Monitor/Display

3 SYSTEM IMPLEMENTATION

3.1 Overview of Development Tool

WEKA Tool

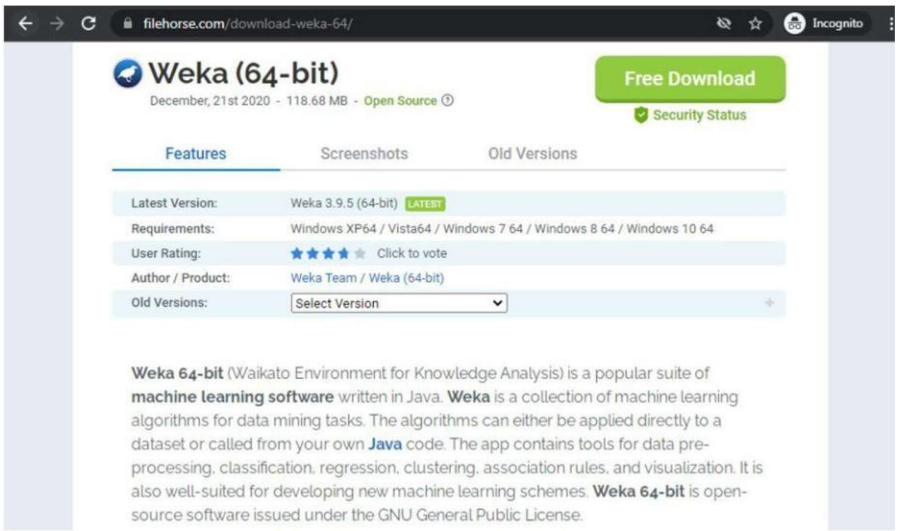
Steps to Install

WEKA Step 1: Download WEKA

Go to the official WEKA website:

https://www.cs.waikato.ac.nz/ml/weka/

- Click on the **Download** section
- Choose the appropriate version based on your operating system



(Windows, macOS, or Linux)

Step 2: Install WEKA

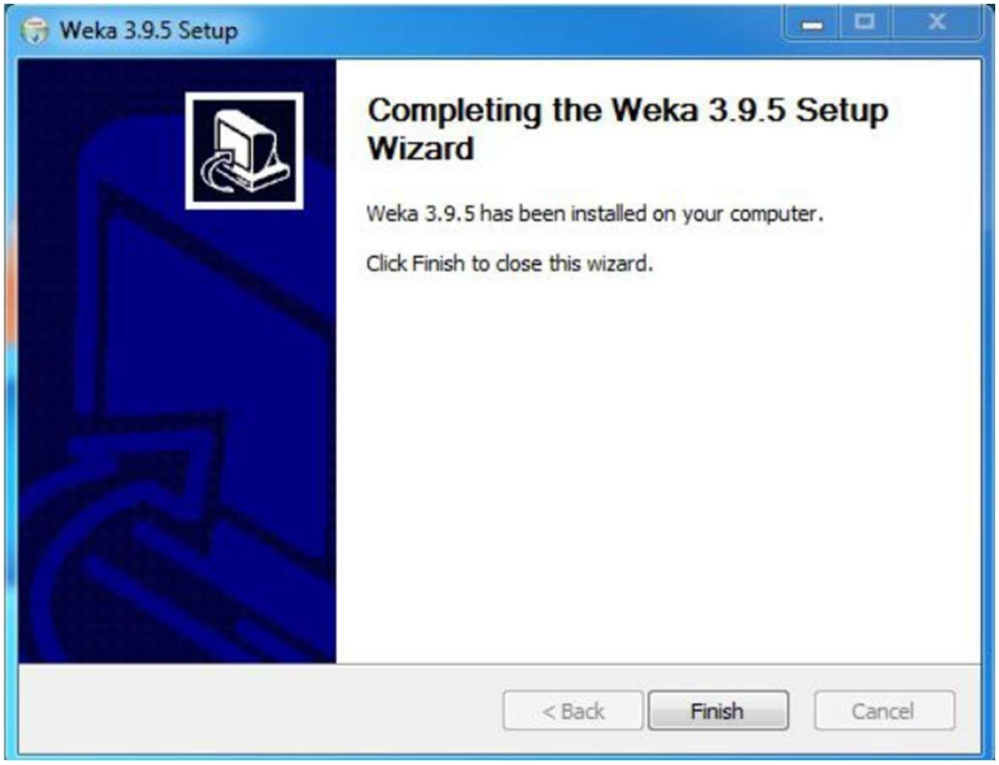
- After downloading, open the installer file

- Follow the installation instructions:
  - Click **Next**

  - Accept the license agreement

  - Choose installation location

  - Click **Install**



Step 3: Launch WEKA

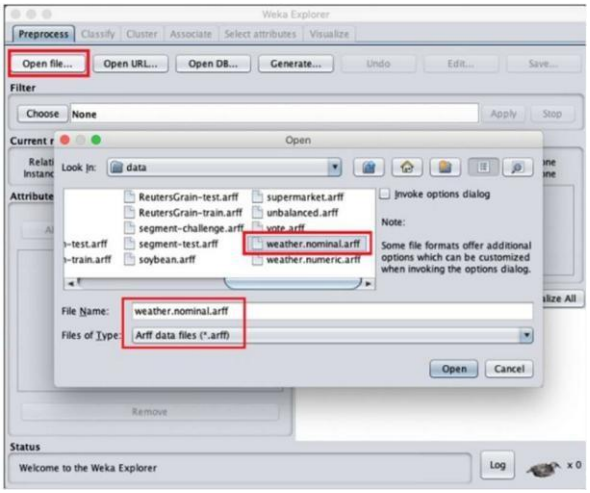
- After installation, open WEKA from:
    - Desktop shortcut OR
    - Start Menu
- The WEKA GUI Chooser will appear with options:
    - Explorer
    - Experimenter
    - Knowledge Flow
    - Simple CLI

Launch WEKA using WEKA Explorer

- Click on **Explorer**
- It contains the main interface for:
    - Loading datasets
    - Preprocessing data
    - Applying algorithms

Loading Dataset in WEKA

- Click on **Open File**
- Select your dataset (CSV or ARFF format)
- Data will be displayed with attributes and values



Applying Algorithms in WEKA

- Go to the **Classify tab** → for classification
- Go to the **Cluster tab** → for clustering
- Go to the **Associate tab** → for association rules

- Select the desired algorithm:
    - J48 (Decision Tree)
    - Random Forest
    - K-Means
- Click **Start** to run the model

4 DATA SET

4.1 Overview

The dataset used in this project is a student attendance dataset containing 105 records, where each record represents an individual student. It includes both academic and attendance-related attributes, which are used to analyze attendance patterns and their impact on student performance using data mining techniques in WEKA.

The dataset contains attributes such as student ID, name, class, section, total marks, and attendance. These attributes provide a combined view of academic performance and student behavior, enabling better analysis compared to using marks alone.

The data is stored in CSV format and organized in a tabular structure, making it easy to preprocess using Python and apply data mining algorithms in WEKA. Through this dataset, relationships between attendance and performance can be identified, and students with low attendance can be detected.

Overall, the dataset plays an important role in understanding student behavior and supports effective analysis and prediction of academic performance.

4.2 Dataset Attributes

The dataset contains the following attributes:

- **Student ID:** A unique identifier assigned to each student
- **Name:** Name of the student
- **Class:** Academic class or grade
- **Section:** Section of the student
- **Total Marks:** Overall marks obtained by the student
- **Attendance:** Attendance percentage or number of days present

These attributes provide both academic and behavioral information, which helps in analyzing student performance more effectively. By considering attendance along with marks, the dataset enables a deeper understanding of student learning patterns.

4.3 Dataset Characteristics

The dataset consists of 105 records, each representing a different student. It is organized in a tabular format, where rows correspond to students and columns represent attributes.

The dataset is stored in CSV (Comma Separated Values) format, which allows easy processing and compatibility with tools like Python and WEKA. It includes both numerical data (such as marks and attendance) and categorical data (such as class and section).

Due to its structured format and balanced attributes, the dataset is suitable for applying various data mining techniques such as classification and clustering for meaningful analysis.

4.4 Dataset Description

This dataset combines academic performance data (total marks) with attendance data, providing a comprehensive view of student behavior and outcomes. It allows for better analysis compared to using only academic scores.

By analyzing this dataset, it becomes possible to:

- Identify the relationship between attendance and academic performance
- Detect students with low attendance and poor performance
- Analyze attendance trends and patterns over time
- Group students based on similar characteristics

Generally, students with higher attendance tend to perform better academically, while irregular attendance may lead to lower performance.

4.5 Role of Dataset in the Project

The dataset plays an important role in different stages of the project:

- Data Preprocessing: Cleaning missing values, removing inconsistencies, and preparing the dataset using Python tools
- Data Visualization: Creating graphs and charts to understand attendance distribution and identify

patterns.

- Data Mining (WEKA):
    - Classification: Predicting student performance or attendance category
    - Clustering: Grouping students based on similar attendance behavior

This step-by-step usage helps in extracting meaningful insights from the dataset.

4.6 Advantages of the Dataset

- Provides a combined view of academic and attendance data
- Helps in early identification of at-risk students
- Supports accurate analysis and prediction
- Easy to use due to its structured CSV format
- Suitable for applying multiple data mining techniques

4.7 Summary

The dataset is a key component of this project, containing 105 student records with both academic and attendance details. It enables effective analysis, pattern identification, and prediction using data mining techniques. The insights obtained from this dataset help educators make informed decisions and improve student performance through better attendance management.

5 SYSTEM ARCHITECTURE

The system architecture of this project represents the overall workflow for analyzing student academic performance using data mining techniques. It consists of multiple stages that transform raw data into meaningful insights through a structured process.

The first stage is **Data Collection**, where student performance data and student activity data are gathered from reliable sources. These datasets include attributes such as marks, grades, attendance, study hours, assignment completion, and participation levels.

The next stage is **Data Integration**, where both datasets are combined using a common identifier (Student ID). This step creates a unified dataset that includes both academic and behavioral information, enabling comprehensive analysis.

The third stage is **Data Preprocessing**, which is performed using Python tools. In this stage, missing values are handled using median and mode imputation, inconsistent or invalid data is removed, and range validation is applied to ensure accuracy. Noise and outliers are also reduced to improve data quality. Data transformation and normalization are carried out to prepare the dataset for further analysis.

After preprocessing, the system moves to the **Data Visualization** stage. Visualization techniques such as histograms and boxplots are used to understand data distribution, detect outliers, and identify trends and patterns.

The next stage is **Data Mining**, where the processed dataset is imported into WEKA. Various algorithms are applied in this stage:

Classification (Decision Tree, Random Forest) to predict student performance
Clustering (K-Means, Hierarchical Clustering) to group students with similar characteristics

Following this, the **Model Evaluation** stage is carried out. The performance of different models is evaluated using metrics such as accuracy, precision, and recall. This helps in selecting the most effective algorithm for prediction.

Finally, the system produces **Results and Insights**, where meaningful conclusions are drawn. The system identifies key factors affecting student performance and provides insights that help educators make data-driven decisions.

[Data Collection]

↓

[Data Integration]

↓

[Data Preprocessing]

↓

[Data Visualization]

↓

[Data Mining (WEKA)]

↓

[Model Evaluation]

↓

[Results & Insights

6 METHODOLOGY AND ALGORITHMS

6.1 Introduction

This chapter explains the data mining techniques and algorithms used in the project to analyze and predict student academic performance. After preprocessing the dataset, various techniques such as classification, clustering, and association rule mining were applied using WEKA. These techniques help in extracting meaningful patterns, grouping students, and predicting performance.

6.2 Classification Techniques

Classification is used to predict the category or class of student performance based on input features such as attendance, study hours, and previous scores.

6.2.1 Decision Tree (J48)

Decision Tree is a supervised learning algorithm used for classification. In WEKA, the J48 algorithm is an implementation of the C4.5 decision tree.

Working:

The dataset is split based on attribute values
Each internal node represents a condition Each
section represents an outcome

Application in Project:

Used to classify students based on performance and identify influencing factors such as attendance and Total marks.

6.2.2 Random Forest

Random Forest is an ensemble learning algorithm that uses multiple decision trees to improve prediction accuracy.

Working:

Multiple decision trees are created using random subsets of data

Each tree gives a prediction

Final output is based on majority voting

Application in Project:

Used to predict student performance with higher accuracy. It performed better compared to other classification models.

6.3 Clustering Techniques

Clustering is an unsupervised learning method used to group students based on similar characteristics.

6.3.1 K-Means Clustering

K-Means is a partition-based clustering algorithm.

Working:

Data is divided into K clusters

Each cluster has a centroid

Data points are assigned to the nearest centroid

Centroids are updated iteratively

Application in Project:

Used to group students into categories such as high-performing, average, and low- performing.

6.3.2 Hierarchical Clustering

Hierarchical clustering creates a tree-like structure of clusters.

Working:

Starts by treating each data point as a separate cluster

Gradually merges clusters based on similarity

Forms a dendrogram (tree structure)

Application in Project:

Used to understand relationships between different student groups.

7 RESULTS AND ANALYSIS

7.1 Introduction

This chapter presents the experimental results obtained from applying various data mining techniques on the preprocessed student dataset. The objective of this analysis is to evaluate the performance of different algorithms and to identify significant patterns influencing student academic performance. The experiments were conducted using the WEKA tool, and the results are interpreted using standard evaluation metrics.

7.2 Experimental Setup

The cleaned and integrated dataset was imported into WEKA for analysis. The following categories of data mining techniques were applied:

- **Classification:** Decision Tree (J48), Random Forest

- **Clustering:** K-Means, Hierarchical Clustering

The dataset included attributes such as marks, attendance, study hours, assignment completion, and participation level, which were used as input features for the models.

7.3 Classification Analysis

7.3.1 Decision Tree (J48)

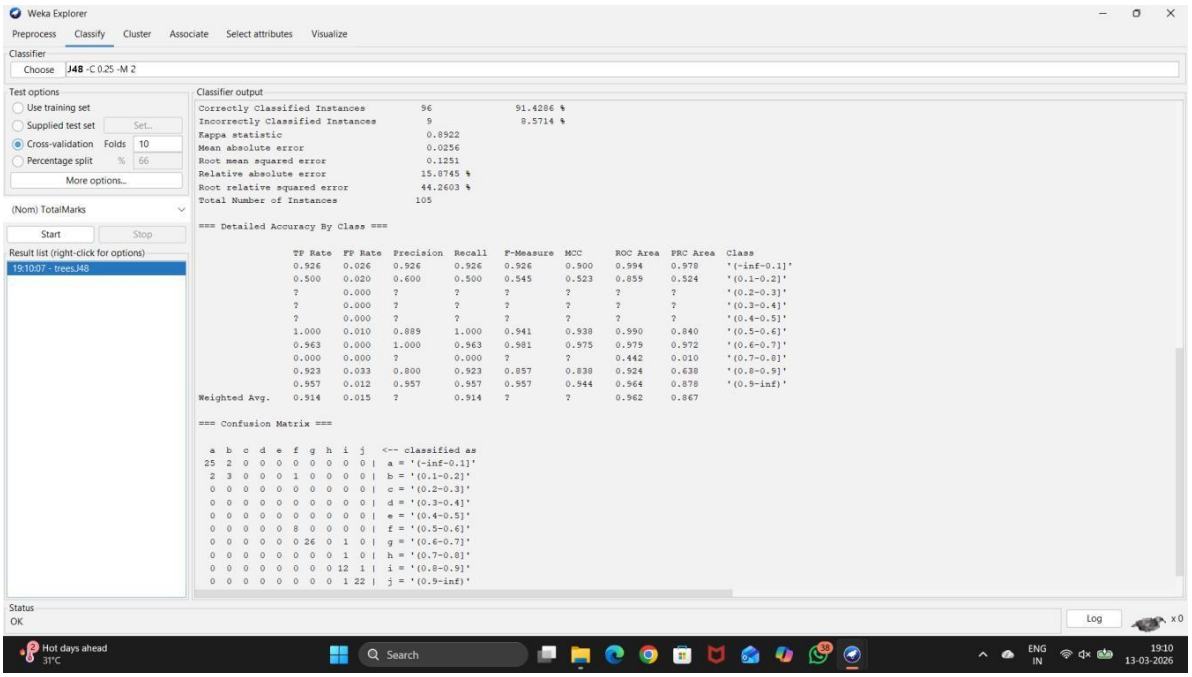
The J48 algorithm generated a tree-based model that classifies student performance based on decision rules derived from input attributes. The model highlighted key attributes such as attendance and study hours as major decision nodes.

Analysis:

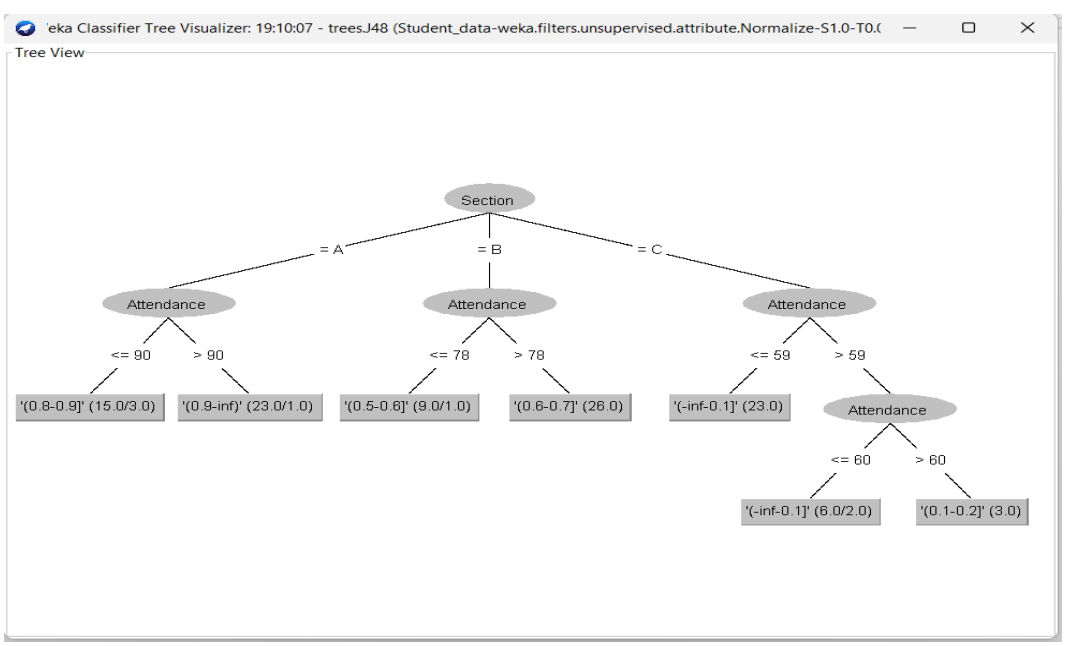
The model is highly interpretable and provides clear decision rules.

It effectively identifies the most influential factors affecting performance.

However, the prediction accuracy is moderate due to possible overfitting and sensitivity to data variations.



Decision Tree / J48



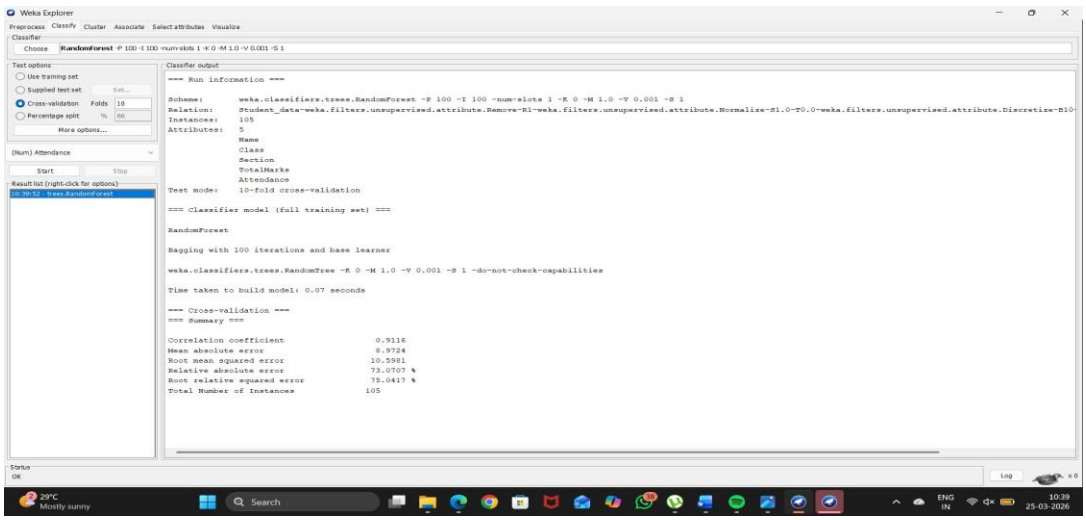
7.3.2 Random Forest

The Random Forest algorithm, being an ensemble method, constructs multiple decision trees and aggregates their predictions to improve accuracy and stability.

Analysis:

Achieved higher accuracy compared to J48.

Reduced overfitting due to multiple tree aggregati



Performance Evaluation

The classification models were evaluated using standard metrics:

- **Accuracy:** Measures overall correctness of predictions
- **Precision:** Measures correctness of positive predictions
- **Recall:** Measures the ability to identify relevant instance

7.4 Clustering Analysis

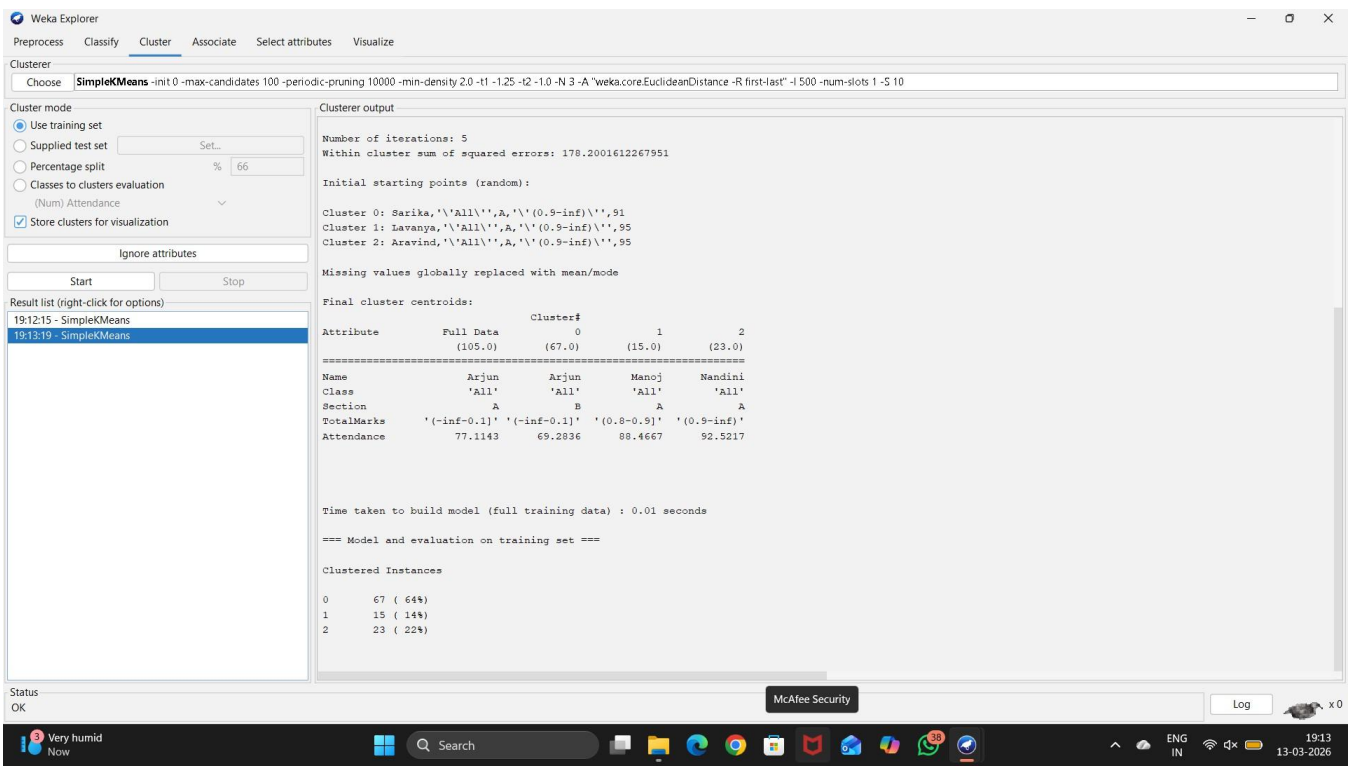
7.4.1 K-Means Clustering

K-Means clustering partitioned the dataset into distinct groups based on similarity in attributes.

Analysis:

Successfully grouped students into categories such as high, medium, and low performers.

Clusters showed strong correlation with attendance and study hours. Provided clear segmentation useful for targeted academic interventions.



7.4.2 Hierarchical Clustering

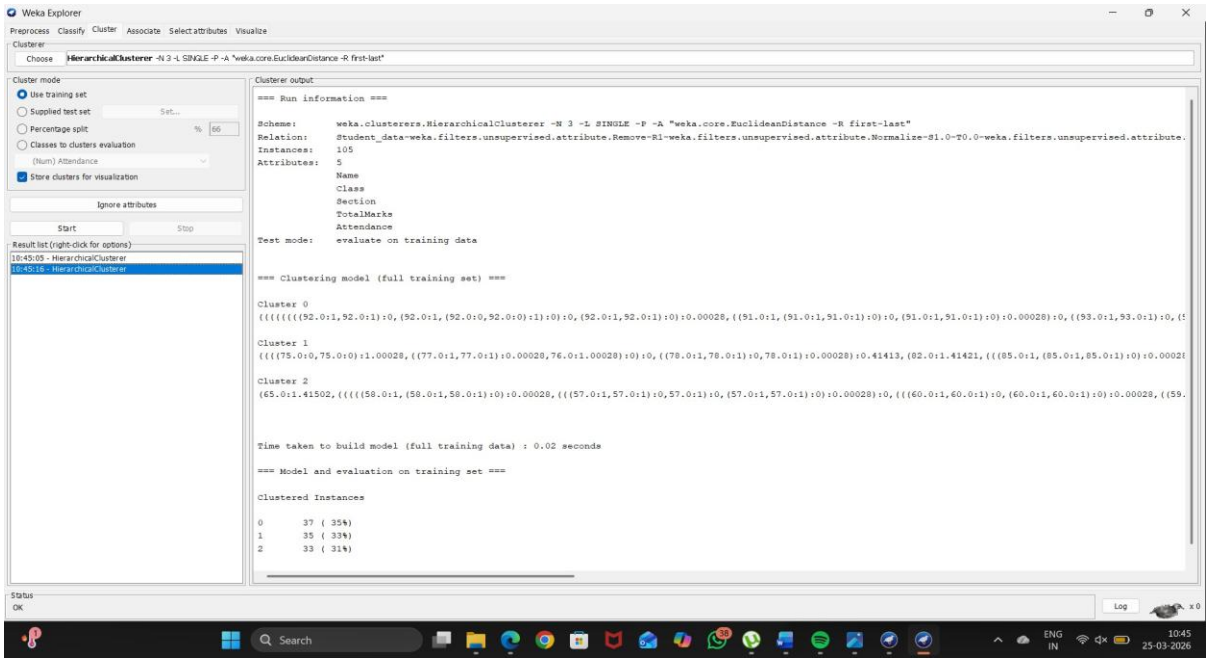
Hierarchical clustering produced a dendrogram representing nested clusters.

Analysis:

Enabled visualization of relationships between student groups.

Did not require predefined cluster count.

Provided deeper insights into the structure of the dataset.



7.5 Key Findings

- Attendance is a critical factor influencing student performance.
- Students with higher attendance tend to perform better.
- Combining academic data significantly improves prediction accuracy.

8 CONCLUSION

This project focused on analyzing **student attendance** using data mining techniques by utilizing a structured attendance dataset. The main objective was to identify patterns, trends, and key factors affecting student attendance, and to develop predictive models that support better academic monitoring and decision-making.

The project began with data collection and preprocessing, where techniques such as handling missing values, removing inconsistencies, and validating data ranges were applied to improve data quality. The cleaned dataset provided a reliable base for analysis. Data visualization techniques were also used to understand attendance distribution, identify irregular patterns, and observe trends among students.

Various data mining techniques were implemented using WEKA, including classification and clustering. Classification algorithms such as Decision Tree (J48) and Random Forest were used to predict attendance categories (e.g., regular and irregular students). Among these, Random Forest produced better accuracy and proved to be more effective. Clustering techniques like K-Means and Hierarchical Clustering were applied to group students based on similar attendance behavior, helping in identifying different attendance patterns.

The analysis showed that attendance is an important factor influencing student performance, and students with low attendance are more likely to face academic difficulties. It also highlighted that early identification of irregular attendance can help educators take timely corrective actions. Additionally, grouping students based on attendance behavior provides better insights into student engagement levels.

Overall, the project demonstrates that applying data mining techniques to attendance data can generate meaningful insights, improve prediction accuracy, and support data-driven decision-making. It also provides a strong foundation for future enhancements such as real-time monitoring systems and integration with academic performance data.

9 FUTURE SCOPE

The project began with data collection and preprocessing, where techniques such as handling missing values, removing inconsistencies, and validating data ranges were applied to improve data quality. The cleaned dataset provided a reliable foundation for further analysis. Data visualization techniques were also used to study attendance distribution, detect irregular patterns, and observe trends among students.

Various data mining techniques were implemented using WEKA, including classification and clustering. Classification algorithms such as Decision Tree (J48) and Random Forest were used to predict attendance categories (e.g., regular and irregular students). Among these, Random Forest achieved better accuracy and proved to be more effective. Clustering techniques like K-Means and Hierarchical Clustering were applied to group students based on similar attendance patterns.

The analysis showed that student attendance is a key factor in identifying learning behavior, and students with low attendance can be considered at risk. Early detection of irregular attendance enables educators to take timely corrective measures. Grouping students based on attendance behavior also provides better insights into participation and engagement levels.

Overall, this project demonstrates that applying data mining techniques to student attendance data can generate valuable insights, improve prediction accuracy, and support data-driven decision-making. It also provides a foundation for future enhancements such as real-time attendance monitoring and advanced analytical models.

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